



Three-dimensional feature parameters measurement of bubbles in gas–liquid two-phase flow based on virtual stereo vision

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ABSTRACT

In this paper, three-dimensional (3D) feature parameters reconstruction and measurement methods of bubbles in gas–liquid two-phase flow based on virtual stereo vision have been developed. A mathematical model of virtual stereo vision is established, and the optimized design method of structure parameters is discussed in detail. In addition, the key technologies for the 3D measurement of bubbles, including the camera calibration, the virtual stereo vision sensor calibration, image processing, feature extraction and 3D reconstruction of bubble trajectories are introduced. All the intrinsic and extrinsic parameters of two virtual cameras, as well as coefficients of lens distortion, can be estimated by the complete mathematical model of the camera. With the sensor calibration, the relative position and orientation of two virtual cameras are determined. Several improved image processing algorithms are used to extract the primary modality parameters of bubbles, and then the 3D trajectories of bubbles can be reconstructed based on the measurement model of virtual stereo vision. The experiment is tested on the upstream bubble-producing appliance. It shows that the method is valid, and the 3D trajectories of bubbles can be reconstructed with high precision. The measured absolute error of bubble 3D displacement is smaller than 0.13 mm, and the relative error is smaller than 0.49%.

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1. Introduction

With the rapid development of science and technology, gas–liquid two-phase flow has been widely used in many fields such as power, chemical, nuclear, petroleum, metallurgy and environmental engineering. The effect on the engineering application largely depends on the bubble modality and the interaction between the discrete phase and the continuous phase [1,2]. In order to better understand the mechanism of gas–liquid two-phase flow and to describe the movement characteristics of the dispersed phase, an accurate 3D measurement for the modality and motion parameters of bubbles is of great significance.

Currently, the measurement methods of bubble feature parameters include the invasive measurement based on the probe [3], Laser Doppler Velocity (LDV) [4], Process Tomography (PT) [5], ultrasonic measurement [6], numerical simulation and so on. The invasive measurement based on the probe can acquire the local information of flow field accurately, but it will inevitably interfere with the flow field and bring measurement error because of chemical corrosion after being used for a long time. LDV has the merits of good linearity, high spatial resolution and rapid response, but it can only observe the individual spot of the flow field rather than

the full field of the non-stable flow. PT can measure the spatial and temporal distribution of the two-phase flow non-intrusively, but it is difficult to reconstruct the image and implement the real time measurement. The numerical simulation method achieves bubble parameters based on the equation of bubble dynamics, so it will be more complicated to compute. With the rapid development of optoelectronics technology, computer vision and image processing, the parameter measurements of the gas–liquid two-phase flow based on high-speed photography [7–9] have made giant strides in development. The high-speed photograph methodology is a full field, non-intrusive, instantaneous measurement for the feature parameters of bubbles in the gas–liquid two-phase flow. Traditionally, in order to obtain 3D flow field data, a binocular stereo vision sensor that contains two high-speed cameras is applied. However, it is unavoidable to introduce measurement error because of non-synchronous cameras, making the measurement system more complicated and costly.

In this paper, a 3D measurement method for the feature parameters of bubbles based on virtual stereo vision is developed. ‘Virtual stereo vision’ is not really composed of two physical cameras, but one camera and the reflective and prismatic optical path to mirror two virtual cameras. Therefore, it does not require a synchronous controller, has rapid measurement speed and with low cost. Kent and Eaton [10] applied the optical setup of stereo photography to study the He-filled bubbles in a transparent engine cylinder, but there was a lack of efficient imaging processing and computing power at that period. It

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Nomenclature

β	Angle of the projection center O' to the image plane ($^\circ$)
f	Focus length of lens (mm)
α	Angle between the reflector $M_1(M_2)$ and the optical axis $O'O$ ($^\circ$)
γ	Angle between the reflector sets M_1, M_2 and P_1, P_2 ($^\circ$)
d	Horizontal distance between the reflector sets M_1, M_2 and P_1, P_2 (mm)
h_3	Vertical distance between the reflector M_1, M_2 and the projection plane of camera (mm)
θ	Angle between the virtual optical axis $O'O$ and z axis ($^\circ$)
h_0	Nearest distance between the valid field of view and the reflector M_1, M_2 (mm)
h_1	Vertical distance between the reflector sets and the projection plane of virtual cameras (mm)
Δh	Depth of the valid field of view (mm)

used a simple calibration method, and ignored several key factors affecting the measurement accuracy, such as structure optimization, measurement model establishment and so on. Bardet et al. [11] used a split-screen single camera to obtain the left and right images of the turbulent confined swirling. Combined with the stereoscopic PIV processing of seeding particles, they identified the free surface of the swirling and analyzed the flow characteristics. Cheung et al. [12] used one CCD video camera and an off-the-shelf stereo adaptor to obtain the digital stereoscopic image. They discussed the variable shutter speeds to control the trajectory length of the tracked particles, and developed a color coding method for matching particle pairs. The same configuration was applied in the reconstruction of the 3D geometry of flame surfaces by Ng and Zhang [13]. They calculated the surface area of the flame based on the 3D coordinates of each pixel, and the error was as small as 8%. Due to the use of the off-the-shelf stereo adaptor, all the structure parameters of the system cannot be optimized further according to the specific measurement. In this paper, for the feature parameters of bubbles in the gas–liquid two-phase flow, we give a complete measurement method. In order to reconstruct the 3D parameters of bubbles, several key steps are implemented. Firstly, all the intrinsic and extrinsic parameters with the coefficient of lens distortion of two virtual cameras are calibrated. Secondly, the relative position and orientation of the two virtual cameras are determined. Then the original bubble image is processed, and the main modality parameters are extracted. After a proper matching algorithm, the 3D trajectory of bubbles can be reconstructed finally. We emphasize more about several key technologies affecting measurement accuracy, such as structural optimization design of the virtual stereo vision sensor, mathematical machinery of the camera and the sensor, improved image processing and auto-identification, and the method of accuracy evaluation, as they are little mentioned in other papers. The experiment is tested on the upstream bubble-producing appliance. It shows that the proposed technologies are valid and with high precision.

The paper is organized as follows. Section 2 describes the measurement principle of virtual stereo vision. Section 3 introduces the mathematical model of virtual stereo vision, and the optimized design of structure parameters. Section 4 develops the complete 3D measurement method for the feature parameters of bubbles, including two virtual cameras calibration, calibration of the virtual

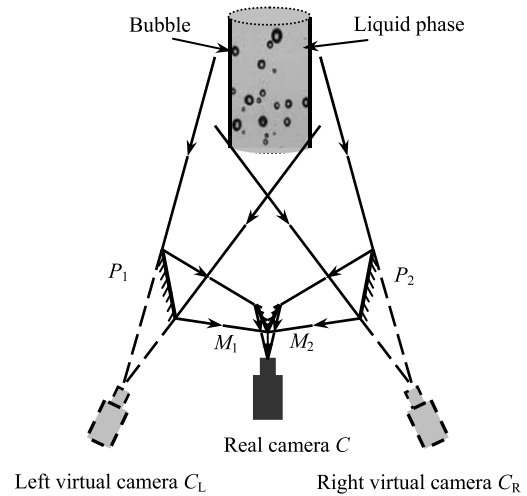


Fig. 1. Measurement principle of virtual stereo vision.

stereo vision sensor, image processing and 3D tracking of bubbles. Section 5 provides the experimental results. Real measurement data are used to validate the proposed method. Finally, we give concluding remarks in Section 6.

2. Measurement principle of the virtual stereo vision

The 3D measurement principle of virtual binocular stereo vision for bubbles is as shown in Fig. 1. The virtual stereo vision sensor consists of one high-speed camera, symmetrical reflector sets M_1, M_2 in the middle and symmetrical reflector sets P_1, P_2 on both sides. Through the refractive and prismatic optical paths, the real camera C can be mirrored as two symmetrical virtual cameras C_L and C_R , whose optical axes are intersecting, and the image plane of the real camera C is separated into the left virtual image plane and the right virtual image plane. When bubbles are observed, the measured features are imaged in the vision of the two virtual cameras to form virtual stereoscopic parallax. That is, the left virtual camera C_L and the right virtual camera C_R act as binocular stereo vision, and we can acquire the stereo image pairs of the measured features on one image plane of the high-speed camera.

Therefore, virtual binocular stereo vision can measure the 3D feature parameters of bubbles just like binocular stereo vision, besides the field of view is reduced by half. For the gas–liquid two-phase flow of the particular dynamic measured object, it does not need two severely synchronous image signals, which decreases the complexity effectively and has relatively rapid speed.

3. Optimum design of the virtual stereo vision sensor

The structure parameters of virtual stereo vision should be optimized for the specific measurement object upon request, because the confluence angle and baseline of stereo vision will greatly influence the valid field of view and the measurement error. Wang et al. [14] analyzed part of the structure parameters and influence factors of the FOV, discussed three different arrangements and made the simulation based on the TracePro software. We establish the mathematical model of the virtual stereo vision, and conduct a comprehensive analysis of all structural parameters.

3.1. Mathematical model of virtual stereo vision

Fig. 2 shows the mathematical model of virtual stereo vision. Note that L is the lens of the real high-speed camera, K is the real image plane of the high-speed camera, $O'O$ is the optical axis of the

$$\begin{cases} x = \frac{(X_1 + X_2)(fd_0 \sin \theta - fs)}{f \cos 2\theta (X_1 - X_2) - (X_1 X_2 + f^2) \sin 2\theta} \\ y = \frac{2Y_1 Y_2 \sin \theta}{f(Y_1 - Y_2)} x \\ z = \frac{f(s \sin 2\theta - d_0 \cos \theta)(X_1 - X_2) + 2X_1 X_2(d_0 - s \sin \theta) \sin \theta + 2f^2 s \cos^2 \theta}{f \cos 2\theta (X_2 - X_1) + (X_1 X_2 + f^2) \sin 2\theta} \end{cases} \quad (1)$$

Box I.

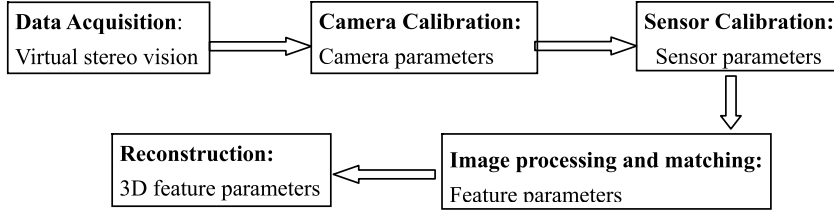


Fig. 3. Key procedure to measure 3D feature parameters of bubbles.

4. Three-dimensional measurement of bubbles

For the gas–liquid two-phase flow of the dynamic measured object, the complete measurement procedure is shown in Fig. 3. The first is to acquire the image pairs based on the optimized virtual stereo vision sensor, then the camera and sensor calibration should be carried out to estimate the intrinsic and extrinsic parameters with lens distortion of the camera, and establish the relationship between the real world and stereo image coordinates. The image processing and matching follows to enhance the image quality, extract the main modality feature parameters and match the feature pairs of bubbles. Finally, the 3D motion parameters of bubbles are reconstructed based on the measurement model of the virtual stereo vision.

4.1. Camera calibration

The calibration of the camera and the virtual stereo vision sensor is very important to provide the level of accuracy in 3D reconstruction. Fig. 4 shows the perspective projection model of the camera in virtual stereo vision. Note that O_u is the center of the camera image plane π_i , O_c is the projection center of the camera and z_c axis is the optical axis of the camera lens. The relative coordinate frames are defined as follows: $O_w - x_w y_w z_w$ is the 3D world coordinate frame, $O_c - x_c y_c z_c$ is the 3D camera coordinate frame and $O_u - X_u Y_u$ is the 2D image plane coordinate frame. Given one point P_w in 3D space, its homogeneous coordinate in $O_w - x_w y_w z_w$ is denoted by $\tilde{P}_w = (x_w \ y_w \ z_w \ 1)^T$. P_u is the ideal projection of P_w in the camera image plane, while P_d is the real projection because of lens distortion. In the same way, we can use $\tilde{P}_u = (x_u \ y_u \ 1)^T$ to denote P_u 's homogeneous coordinate in $O_u - X_u Y_u$.

According to the pinhole perspective projection theory of the camera, the relationship between a 3D point and its image projection can be given by

$$\tilde{s}P_u = A \begin{bmatrix} R & T \end{bmatrix} \tilde{P}_w \quad (11)$$

where $A = \begin{bmatrix} f_x & 0 & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}$ is called the camera intrinsic parameter matrix, with (u_0, v_0) the coordinates of the principal point, f_x, f_y the effective focal length in pixels in the x and y directions respectively. s is an arbitrary scale factor. The camera extrinsic parameter R is a 3×3 orthogonal rotation matrix, and T is a 3×1 translation vector.

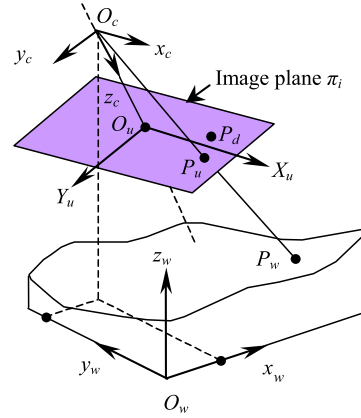


Fig. 4. Perspective projection model of the camera in virtual stereo vision.

To achieve 3D measurement of bubbles with high precision, complex lens distortion should be considered. The radial lens distortion can be given by

$$\delta \begin{bmatrix} u^r \\ v^r \end{bmatrix} = k^r \begin{bmatrix} x_u \\ y_u \end{bmatrix}, \quad k^r = k_1 r^2 + k_2 r^4 + \dots \quad (12)$$

where $r^2 = x_u^2 + y_u^2$, k_1 and k_2 are the coefficients of the radial distortion. The tangential lens distortion is given by

$$\delta \begin{bmatrix} u^t \\ v^t \end{bmatrix} = \begin{bmatrix} 2p_1 x_u y_u + p_2 (r^2 + 2x_u^2) \\ p_1 (r^2 + 2y_u^2) + 2p_2 x_u y_u \end{bmatrix} \quad (13)$$

where p_1 and p_2 are the coefficients of tangential distortion. Thus actual projection coordinate can be represented as follows:

$$\begin{bmatrix} x_d \\ y_d \end{bmatrix} = \begin{bmatrix} x_u + \delta u^r + \delta u^t \\ y_u + \delta v^r + \delta v^t \end{bmatrix}. \quad (14)$$

Eqs. (11)–(14) completely describe the real perspective projection model of the camera in the virtual stereo vision sensor.

Due to the nonlinear nature of the camera model with lens distortion, the estimated coordinate and the real extraction of feature points is often inconsistent. Thus an iterative optimizing algorithm is used to minimize the sum of squared residuals between the model and observations. The optimizing objective function is expressed by

$$F(x) = \sum_{i=0}^n ((U_i - u_i)^2 + (V_i - v_i)^2) \quad (15)$$

where (u_i, v_i) is the real extracted image coordinate, and (U_i, V_i)

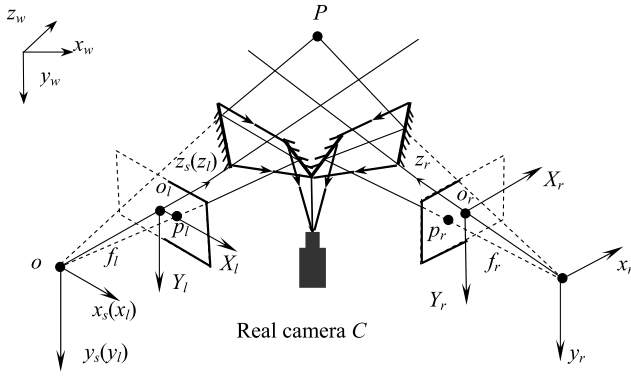


Fig. 5. 3D measurement model of virtual stereo vision.

is the estimated image coordinate based on the camera model. Therefore, all the intrinsic and extrinsic parameters with lens distortion coefficients can be estimated through the Levenberg–Marquardt least squares fitting algorithm [15].

4.2. Calibration of the virtual stereo vision sensor

Fig. 5 shows the 3D measurement model of the virtual stereo vision sensor, which relates the left camera coordinate frame to the right camera coordinate frame. The relative coordinate frames are defined as follows: $O_w x_w y_w z_w$ is the 3D world coordinate frame, $O_l x_l y_l z_l$ and $O_r x_r y_r z_r$ are the 3D coordinate frame of the left virtual camera and its responding right half image plane coordinate frame, and $O_s x_s y_s z_s$ is the sensor measurement coordinate frame, which overlaps with $O_l x_l y_l z_l$.

Given an arbitrary 3D point P , let $\tilde{P}_s = [x_s \ y_s \ z_s \ 1]^T$ be the 3D homogeneous coordinate of point P in $O_s x_s y_s z_s$, $P_r = [x_r \ y_r \ z_r \ 1]^T$ be the 3D coordinate in $O_r x_r y_r z_r$, p_l, p_r are the ideal projections of P in the left and right half image plane of the camera, their corresponding homogeneous coordinates in $O_l x_l y_l z_l$ and $O_r x_r y_r z_r$ are denoted by $\tilde{p}_l = [X_l \ Y_l \ 1]^T$ and $\tilde{p}_r = [X_r \ Y_r \ 1]^T$. The position and orientation of the two virtual cameras can be described by

$$P_r = M_{sr} \tilde{P}_s = [R \ T] \tilde{P}_s \quad (16)$$

where M_{sr} is the 4×4 transform matrix, $T = [t_x \ t_y \ t_z]^T$ is the translation matrix, and R is the orthogonal rotation matrix between $O_s x_s y_s z_s$ and $O_r x_r y_r z_r$. Based on the perspective projection model of the camera, we have $\rho_l \tilde{p}_l = P_s$ with $\rho_l = z_s$, $\rho_r \tilde{p}_r = P_r$. Therefore, the 3D measurement model of the virtual stereo vision can be represented by

$$\begin{cases} x_s = z_s X_l \\ y_s = z_s Y_l \\ z_s = \frac{t_x - X_r t_z}{X_r (r_7 X_l + r_8 Y_l + r_9) - (r_1 X_l + r_2 Y_l + r_3)} \\ = \frac{t_y - Y_r t_z}{Y_r (r_7 X_l + r_8 Y_l + r_9) - (r_4 X_l + r_5 Y_l + r_6)} \end{cases} \quad (17)$$

The accurate parameters can be estimated by the Levenberg–Marquardt nonlinear iterative optimizing algorithm and the accurate distance of feature points. From Eq. (17), the 3D coordinates of measured feature points can be acquired if their homogeneous image coordinates and R, T are known exactly.

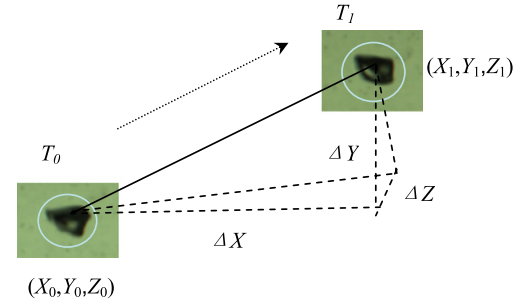


Fig. 6. Scheme of bubble velocity measurement.

4.3. Image processing and feature extraction

The actual bubble images taken by the virtual stereo vision sensor are true color images. In order to extract the useful modality features, such as the perimeter, area, circularity and mass center coordinates, several image processing steps must be performed. Firstly, it is necessary to pre-process with image gray processing, image difference and median filtering. According to the characteristics of bubble images, the improved Otsu method with dynamic threshold compression is used for binary image binarization subsequently. Then the scan line seed filling algorithm is combined with the image morphological processing to fill the bubble holes. In order to eliminate the interference of impurities and noises, the automatic recognition method based on the edge and area attributes [16] is proposed to identify the bubble feature regions. Finally, the useful modality parameters of bubbles can be extracted with sub-pixel precision. For example, the center coordinates $(\bar{x}, \bar{y})$ of a bubble are given by

$$\begin{cases} \bar{x} = M_{10}/M_{00} \\ \bar{y} = M_{01}/M_{00} \end{cases} \quad (18)$$

where M_{10} and M_{01} are the totals of coordinates x, y of bubble pixels respectively, while M_{00} (zero-order moment) denotes the bubble area. The mass center coordinate can be used to reconstruct the 3D motion parameters of bubbles afterwards.

4.4. Three-dimensional reconstruction of motion parameters

The main motion features of bubbles include the trajectory and velocity. Because the size and shape of an air bubble will vary at different times due to the interaction with the liquid phase, we use the epipolar constraint based on the extracted centroid of bubbles, supplemented by the constraints of bubbles' height and projection characteristics, to identify the bubble pairs in the left and right half image accurately. According to the 3D measurement model of virtual stereo vision, if the mass center coordinates of the homonymy bubbles are extracted accurately, and the parameters of the camera and sensor are calibrated with high precision, the 3D position of bubbles can be estimated accurately. Thus the 3D trajectory can be reconstructed by the images at different times.

Fig. 6 shows the measurement principle of 3D velocity. (X_0, Y_0, Z_0) and (X_1, Y_1, Z_1) denote the center coordinates of the same bubble at T_0, T_1 times, while $\Delta X, \Delta Y$ are the displacements in horizontal and vertical directions respectively. The two adjacent frames perform the same operation, extracting the centroid coordinates of a bubble, calculating the 3D position by homonymy matching from the two views, and reconstructing the velocities by motion matching of different frames. The bubble velocity defined as the displacement per unit time is given by

$$v = \frac{\sqrt{(X_1 - X_0)^2 + (Y_1 - Y_0)^2 + (Z_1 - Z_0)^2}}{T_1 - T_0} \quad (19)$$

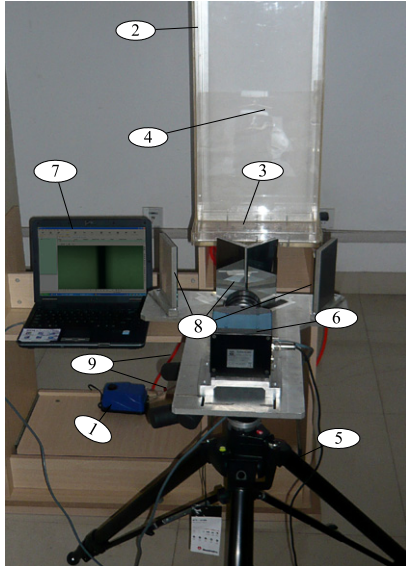


Fig. 7. Experimental set-up. 1-air compressor, 2-background illumination (LED array), 3-air regulator, 4-water tank, 5-tripod, 6-high-speed camera, 7-computer, 8-virtual stereo vision sensor, 9-air pipe.

5. Experiments

5.1. Experimental set-up

The experimental set-up is shown in Fig. 7, which mainly includes the bubble-producing appliance (a transparent rectangular water tank with air compression device), virtual stereo vision sensor, lighting and soft processing platform. The liquid phase in the experiment is the tap water, and the gas phase is the compressed air that is injected into the tank through the air compressor, regulators, easy connectors and the air distributor in turn. The virtual stereo vision sensor consists of a high-speed camera (SpeedCAM MiniVis E2) with 35 mm Nikkor lens and two optical reflector sets. We place the virtual stereo vision sensor in front of the water tank, and the LED array illuminates the bubbles from the back of the tank. The original images (960×720) are captured at 280 frames per second (fps), and Fig. 8 shows the image sequences of three bubbles at intervals of 12 frames. The size of the tested bubbles is about 3–5 mm due to the different diameter of gas orifice and interaction with the liquid phase as the bubbles rise.

5.2. Results

The virtual stereo vision sensor should be calibrated before it is used to measure the 3D feature parameters, chiefly for the 3D trajectories of bubbles. Images of the stereo reference pattern were taken, and the sub-pixel centers of sphere projection were detected. According to the perspective projection model of the camera in 4.1, the left virtual camera parameters are

$$\begin{aligned} f_{xl} &= 3132.6098 \text{ pixels}, & f_{yl} &= 3095.1281 \text{ pixels}, \\ u_{0l} &= 478.3537 \text{ pixels}, & v_{0l} &= 361.2567 \text{ pixels} \\ k_{1l} &= -0.000382, & k_{2l} &= 0.000007, \\ p_{1l} &= 0.000116, & p_{2l} &= -0.000496. \end{aligned}$$

The right virtual camera parameters are

$$\begin{aligned} f_{xr} &= 2989.6931 \text{ pixels}, & f_{yr} &= 3031.8553 \text{ pixels}, \\ u_{0r} &= 458.1328 \text{ pixels}, & v_{0r} &= 360.0713 \text{ pixels} \\ k_{1r} &= 0.001095, & k_{2r} &= -0.000009, \\ p_{1r} &= -0.000026, & p_{2r} &= -0.001837. \end{aligned}$$

According to the mathematical model of the virtual stereo vision sensor in 4.2, the relative position and orientation of two virtual cameras can be represented by

$$R = \begin{bmatrix} 0.359513 & 0.021849 & -0.932884 \\ -0.023320 & 0.999624 & 0.014426 \\ 0.932849 & 0.016568 & 0.359887 \end{bmatrix},$$

$$T = [421.163155 \quad -5.825792 \quad 310.877258]^T.$$

In addition, the 2D modality parameters of bubbles are extracted. According to the image processing method in Section 4.3, the key result is shown in Fig. 9, which is illustrated by an image at $t = 0.257$ s in Fig. 8. Fig. 9(a) shows the image after pre-processing, including image gray processing, image difference and median filtering. The binary image by the improved Otsu method and dynamic threshold compression is shown in Fig. 9(b). From Fig. 9(c), it is can be seen that the bubble holes are filled and the noises are eliminated effectively through auto-recognition of feature regions. The main modality parameters can be calculated by edge extraction, including the bubbles centroid indicated by a cross, as shown in Fig. 9(d).

According to the measurement model of the virtual stereo vision described in Section 4.2, if a series of centroid coordinates of rising bubbles are obtained, and the calibration parameters of the camera and the virtual stereo vision sensor are estimated, the 3D trajectories of three bubbles in Fig. 8 can be reconstructed accurately, which are shown in Fig. 10.

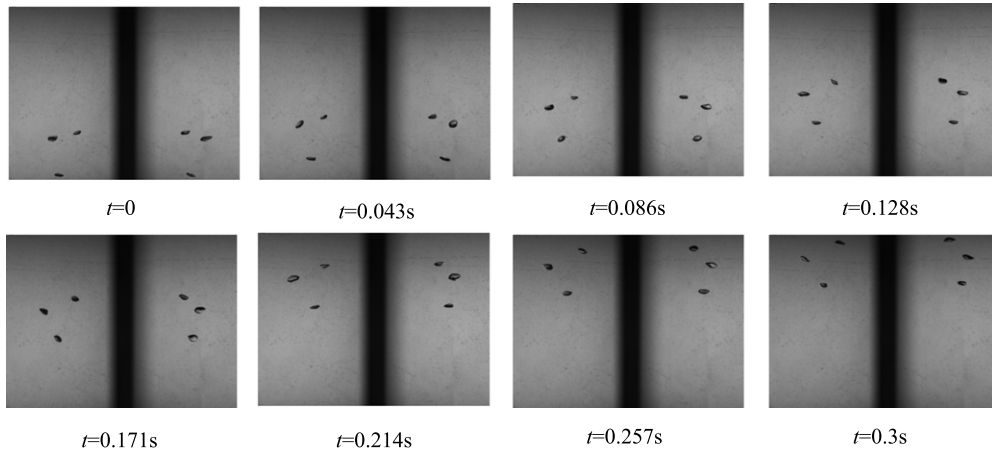


Fig. 8. The original image sequences.

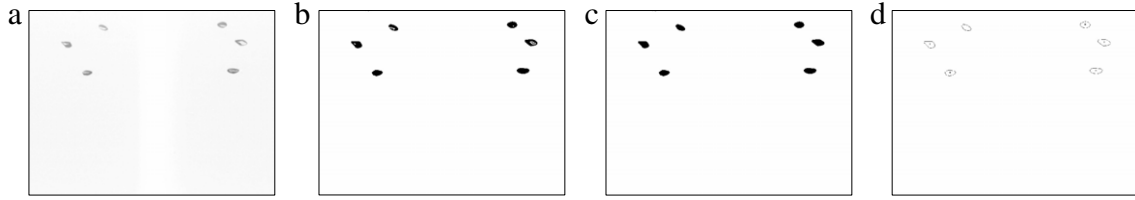


Fig. 9. The image processing result. (a) Image after pre-processing. (b) Image after threshold. (c) Image after hole filling and feature region identification. (d) Image after edge extraction and bubbles centroid (indicated by cross).

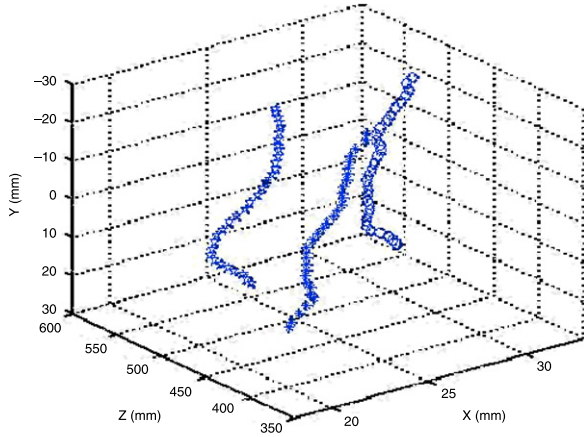


Fig. 10. Three-dimensional trajectories of bubbles.

5.3. Accuracy evaluation

In order to verify the measurement accuracy of the bubble motion parameters, we use multi-spheres whose spatial distance between two arbitrary spheres are known precisely to simulate the bubbles distribution. The spheres can ensure the projection consistency in all directions, so it avoids bringing the measurement error caused by the perspective projection. The multi-sphere template is about 90 mm (L) $\times$ 80 mm (W), which can occupy the field of view basically, and the diameter of the spheres is 5.9 mm. The 9 spheres show a three-dimensional distribution in space. To a certain extent, it can simulate the bubbles at one moment. Fig. 11(a) shows the verifying image captured by the virtual stereo vision sensor. According to the above measurement method, 3D position and displacement of “bubbles” at one moment can be estimated, and compared with the ground truth inspected by a Coordinate Measuring Machine (CMM) with μm -level accuracy. As shown in Fig. 11(b), the horizontal axis represents the number

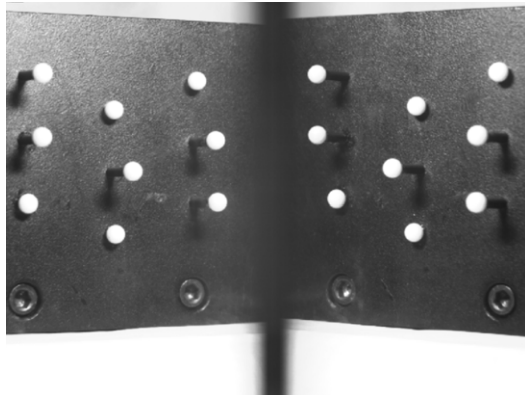
of measured distances between two arbitrary spheres, and the vertical axis represents the measurement error compared with the true value. It shows that the 3D location of bubbles can be reconstructed accurately. The measured absolute error for bubbles 3D position and trajectory is smaller than 0.13 mm and the relative error is smaller than 0.49%, and the proposed method is valid and with high precision.

6. Conclusions

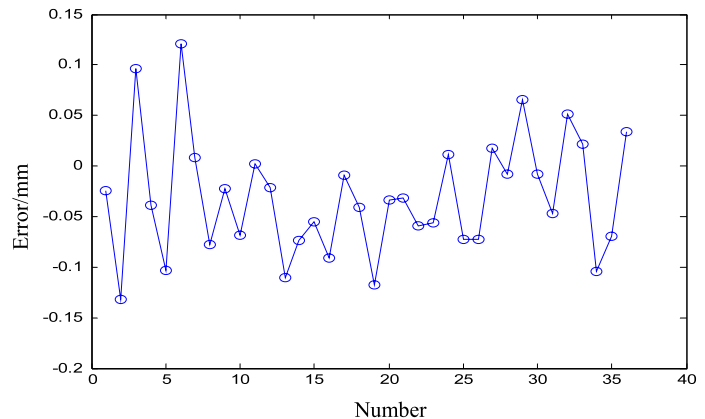
In this paper, a complete method for the 3D reconstruction of bubbles in the gas–liquid two-phase flow based on virtual stereo vision has been developed. Compared with the conventional measurement system based on binocular stereo vision, it avoids the measurement error caused by the non-synchronization of two high-speed cameras, reduces cost and simplifies the calibration. The mathematical model of virtual stereo vision has been established, and the optimum design method of structure parameters has been studied in detail. In addition, the key technologies for the 3D feature parameters measurement of bubbles, including the camera calibration, calibration of the virtual stereo vision sensor, image processing and feature extraction, and motion parameter reconstruction have been introduced. The experiment is tested on the upstream bubble-producing appliance. It shows that the method is valid and with high precision, with the measured absolute error for bubbles 3D position smaller than 0.13 mm, and the relative error smaller than 0.49%.

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(a) Verifying image.



(b) Measurement error.

Fig. 11. The 3D location measurement accuracy of bubbles.

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